AI PORTFOLIO ARTIFACT #2

Demystifying Neural Networks: Structural Architecture & Visual Mechanics

A comprehensive breakdown of key deep learning components and interactive insights from the TensorFlow Neural Network Playground.

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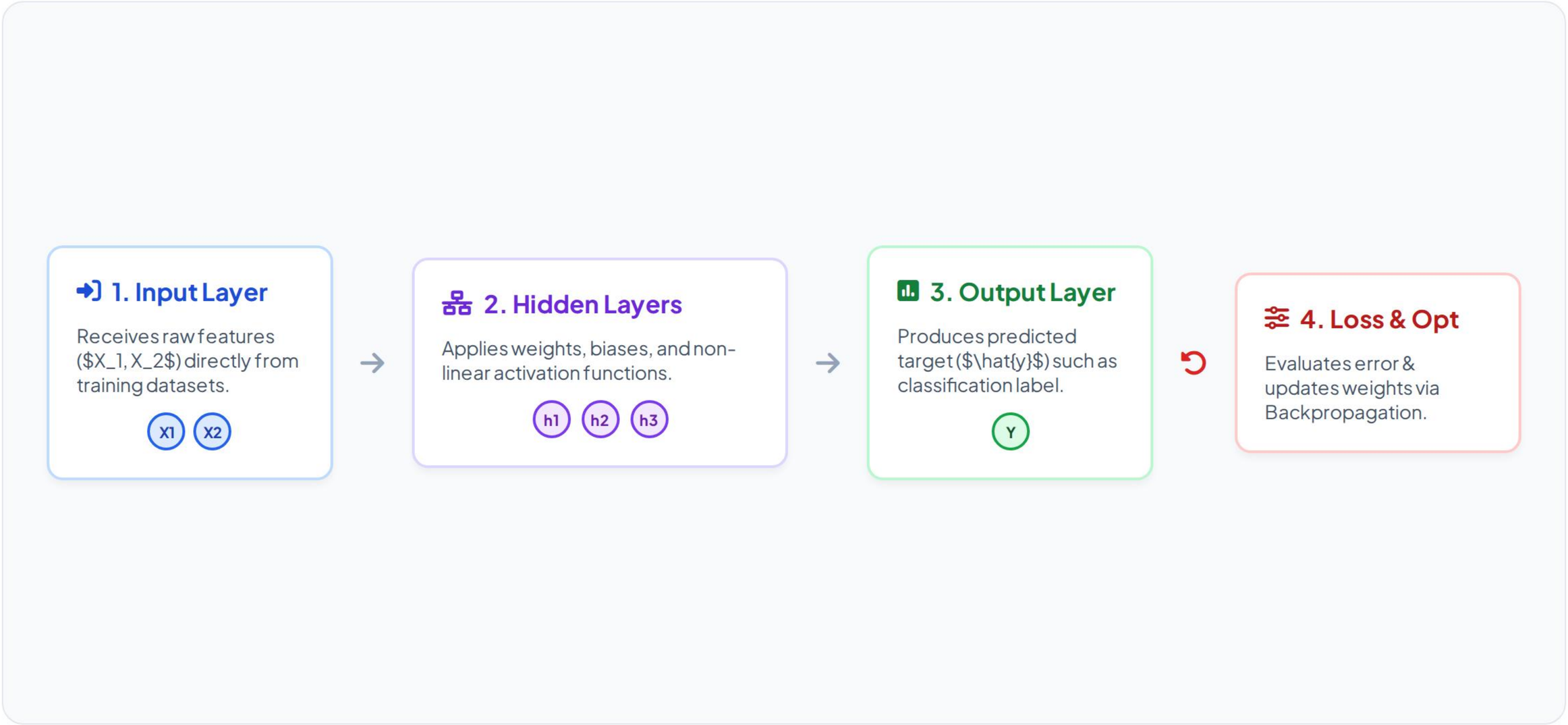
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GLOBAL ARCHITECTURE

End-to-End Neural Network Data Flow

Feedforward Pipeline



CORE ELEMENTS (PART 1)

1. Structural Layers & 2. Artificial Neurons

Neural Layers

Organizational structures containing node groups that extract features at varying abstraction levels.

- **Input Layer:** Passes raw data attributes.
- **Hidden Layers:** Builds feature abstractions.
- **Output Layer:** Yields final predictions.

Artificial Neurons (Nodes)

The fundamental computational unit responsible for aggregating signals and generating output values.

- **Weighted Sum:** $\sum (w_i \cdot x_i) + b$
- **Bias Term:** Shifts activation thresholds.
- **Decision Unit:** Mimics biological firing.

CORE ELEMENTS (PART 2)

3. Connection Weights & 4. Activation Functions

Synaptic Weights (W)

Adjustable parameters representing connection strength between neurons.

- Positive weights amplify input signals.
- Negative weights suppress input signals.
- Learned iteratively during model training.

Activation Functions

Introduces non-linearity enabling networks to learn complex boundary curves.

- **ReLU:** $f(x) = \max(0, x)$ (Fast & sparse)
- **Sigmoid:** Squashes output between 0 and 1.
- **Tanh:** Zero-centered outputs from -1 to 1.

CORE ELEMENTS (PART 3)

5. Loss Functions & 6. Optimization Algorithms

Loss Functions (Cost)

Mathematical metric quantifying disagreement between predictions ($\hat{y}$) and true labels (y).

- **MSE:** Used for continuous regression tasks.
- **Cross-Entropy:** Standard for classification.
- Guides weight adjustment magnitude.

Optimization Algorithms

Algorithms adjusting weights to minimize total loss via gradient descent.

- **Gradient Descent:** Updates weights via gradients.
- **Adam / RMSProp:** Adaptive learning rates.
- **Learning Rate (η):** Step size per epoch.

HANDS-ON EXPERIMENTATION

Neural Network Playground Exploration

Key Empirical Findings

Spiral Dataset

Non-Linear Complexity

Linear activation failed completely. Switching to **ReLU with 3 hidden layers** successfully enclosed the dual spiral boundary.

Noise Impact

Overfitting Risk

Adding 30% noise caused complex models to overfit training points. **L2 Regularization** smoothed decision boundaries.

Learning Rate

Convergence Dynamics

High learning rate (>0.3) caused loss oscillations. A rate of **0.03** delivered smooth, rapid convergence.

CONCLUSION & SYNTHESIS

Value of Visualization & Portfolio Reflection

Intuition Over Black-Box Models

Visual tools demystify neural networks, converting abstract calculus into intuitive visual adjustments of decision boundaries.

Architecture Optimization Trade-Offs

Deeper networks offer higher expressiveness but require careful regularization and tuning to prevent severe overfitting.

Portfolio Breadth Expansion

Complements previous vector embedding analysis by showcasing structural machine learning knowledge and technical communication skills.



Thank You for Reviewing!

Artifact #2 successfully integrated into professional portfolio.